

Digital Logic

BCA — Semester I — 2023 — Answer Sheet
Subject Code: CACS103

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Convert +64.35 into IEEE 754 double precision floating point binary number.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the universal gates with their logic symbol, logic expression, and truth table. Realize an OR Gate using NAND Gate.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Minimize the given Boolean function in both SOP and POS using K-map and realize using basic gates: $F(A,B,C,D) = \Sigma(0,1,2,5,7,8,9,10,13,15)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain 4-bit Binary Parallel Adder. Implement a full Adder using two half Adders.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define combinational circuit. Design a binary to octal converter.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is a synchronous sequential circuit? Explain RS NAND latch with block diagram, logic diagram, truth table, and timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define register. Explain serial-in serial-out register with working mechanism for data input 11010

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Explain ring counter. Design a 3-bit asynchronous up counter with block diagram, count sequence table, and timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define ECL. Explain 3-bit even parity generator and checker.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain the design procedure of a clocked sequential circuit. Design a clocked sequential circuit whose state diagram is given in the figure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.